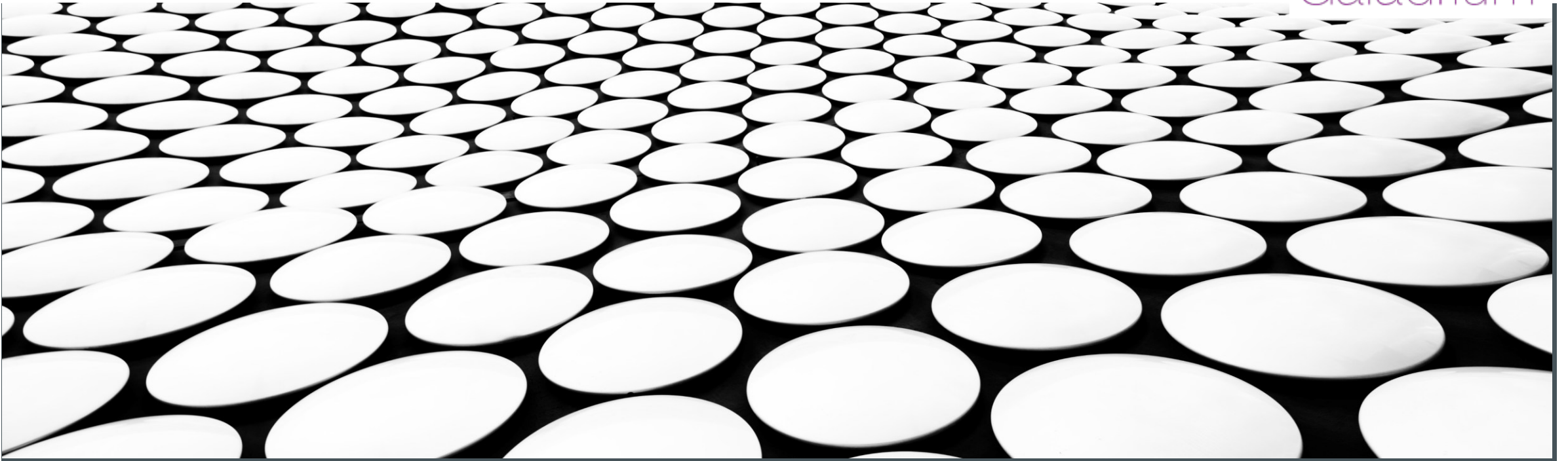


ATTECH TRAINING: AMHS/SWIM GATEWAY INTRODUCTION TO THE COURSE

AMHS/SWIM GATEWAY: MODULE 1

Galadrium

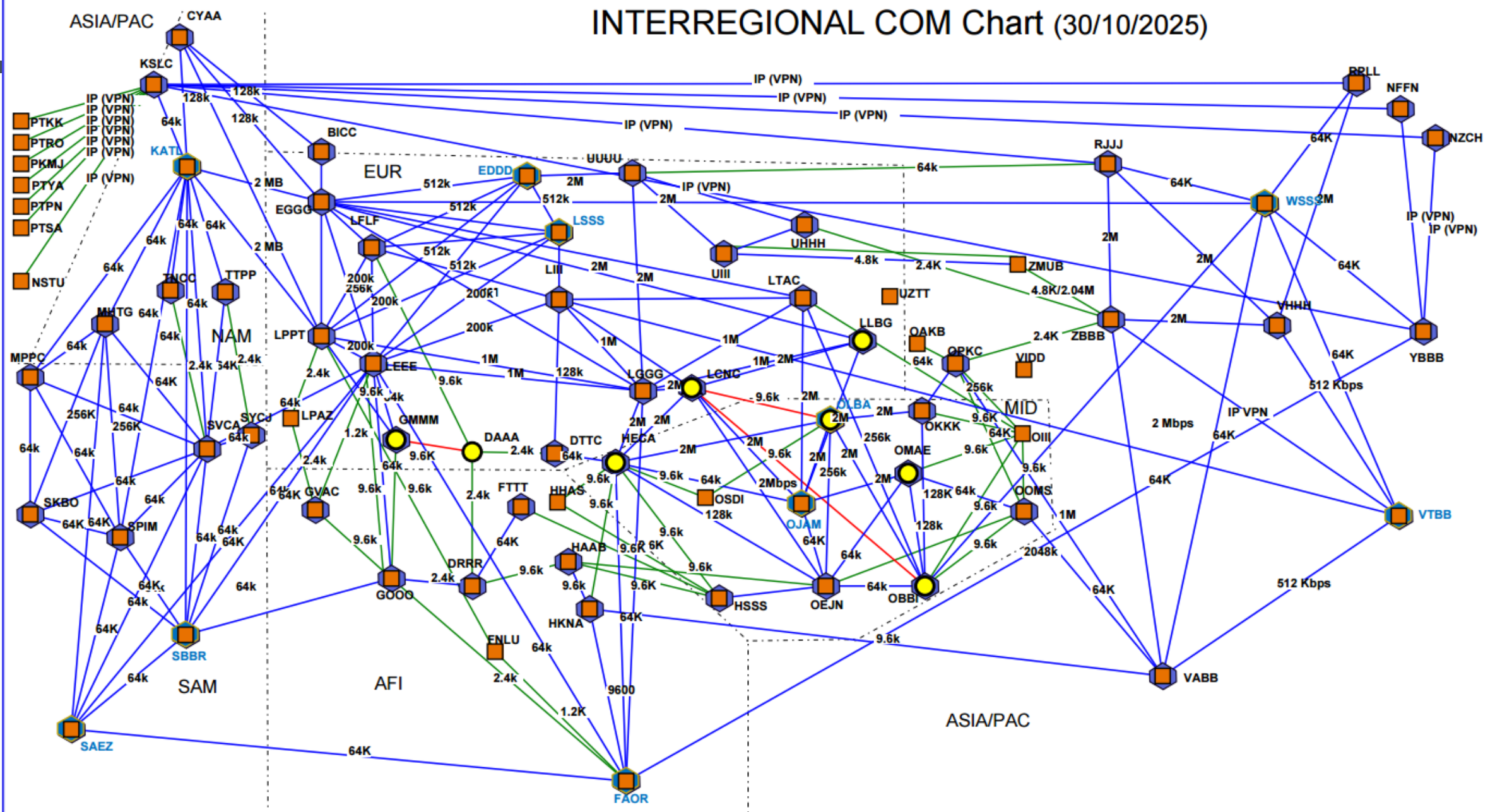


Current Status of ATS communication
protocols worldwide

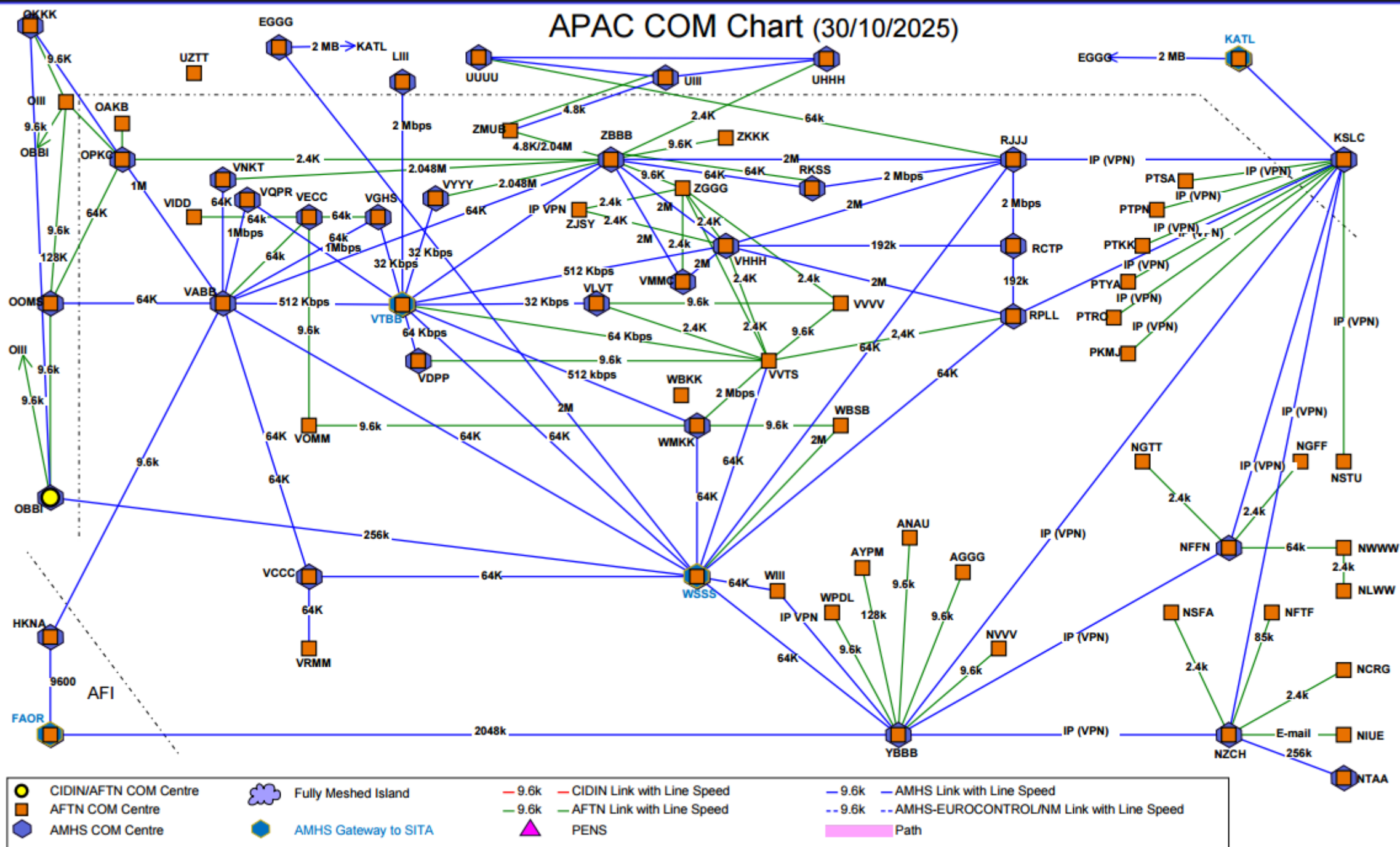
CURRENT STATUS OF ATS MESSAGING

- Currently, two main protocols are used to carry ATS messaging: AFTN and AMHS
- AFTN was introduced in the 1950s
- AMHS was introduced in the 1990s, with the goal of replacing AFTN
- AMHS is today the main communication network for ATS messaging internationally
- AFTN is still being used in parts of the world internationally, and still used a lot domestically
- The SWIM concept was introduced 1997

INTERREGIONAL COM Chart (30/10/2025)



APAC COM Chart (30/10/2025)



CURRENT STATUS OF ATS MESSAGING

- This training is for participants that are already very familiar with AFTN and AMHS
- We won't cover AFTN or AMHS, and we will concentrate on SWIM and the AMHS/SWIM Gateway
- In same areas, to highlight the differences, we will mention AMHS characteristics
- Since SWIM is such a vast new technology, we will only cover some of the most important aspects of it

What is SWIM?

WHAT IS SWIM?

- The SWIM concept (System Wide Information Management) was introduced by Eurocontrol in 1997 to the FAA
- The SWIM concept was adopted by ICAO in 2005
- Is it a foundational technical concept for a modern Air Traffic Network
- It aims to improve information sharing and interoperability between aviation systems
- SWIM aims to be the answer to achieving global interoperability and service orientation in air traffic management
- SWIM core services will enable systems to request and receive information when they need it
- Systems will subscribe for automatic receipt, and publish information and services as appropriate

SWIM STANDARDS

- **ICAO Doc 10039**
Manual on the System-wide Information Management (SWIM) Concept, 1st Edition, 2024
- **ICAO Doc 10203**
Manual on the System-Wide Information Management (SWIM) Implementation, 1st Edition, 2024

Maybe also consider:

- **ICAO Doc 10199** : Procedures for Air Navigation Services-Information Management, 1st Edition, 2024
- **ICAO Doc 9854**: Global ATM Operational Concept
- **ICAO Doc 9882**: Manual on Air Traffic Management System Requirements



SWIM STANDARDS: ICAO DOC 10039

ICAO Doc 10039

- I have this document
- Even though the date of the First Edition is 2024, the content has been worked on for a long time
- This is a must read for anyone working on SWIM
- The definitions of the acronyms are very useful for someone starting on SWIM



SWIM OBJECTIVES

According to EUROCONTROL, SWIM enables:

- Global aviation information exchange and future collaborative processes
- The provision of the right information, to the right people at the right time
- The modernisation and digitalisation of the networked global air navigation system.



SWIM BENEFITS

According to EUROCONTROL, SWIM provides these benefits:

- From an information service consumer viewpoint, SWIM removes data silos and fosters data integration.
- The consistent use of data and preservation of meaning when information is exchanged improves collaborative decision making and shared situational awareness.
- The increased levels of interoperability lower maintenance costs and vendor lock-in.

TBO AND SWIM

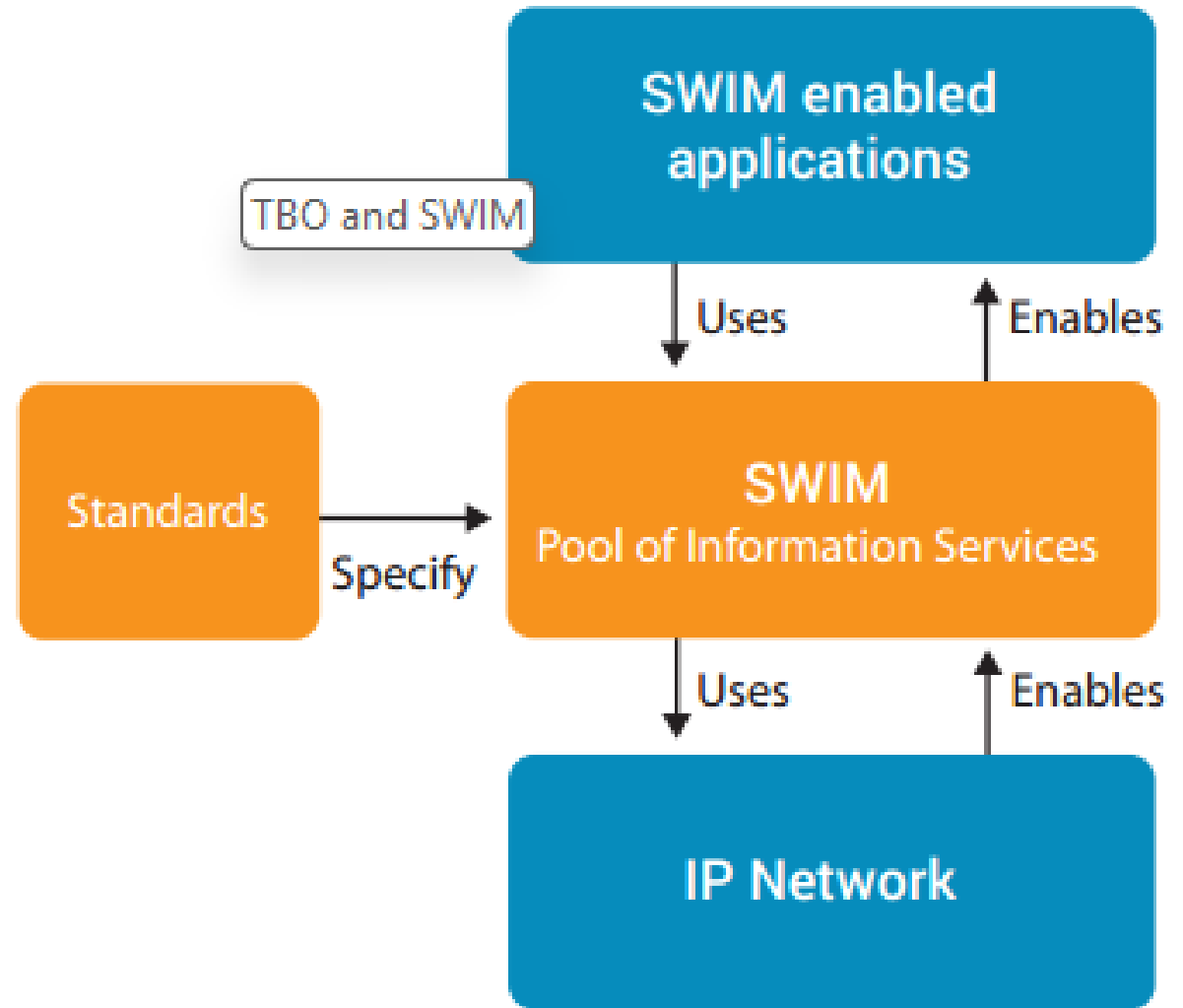
- TBO stands for “Trajectory Based Operations”
- TBO is for sharing and managing trajectories and associated information
- TBO enables the accurate planning, prediction, sharing, and management of an aircraft's 4D trajectory
- The 4D trajectory comprises latitude, longitude, altitude, and time
- The shift to TBO is driven by the need for greater efficiency, capacity, and predictability
- TBO will bring important benefits to the ATM
- But TBO depends on the seamless exchange of aeronautical, weather and flight information
- The information needs to be in accordance with the standards and practices of SWIM
- TBO is just one of the services that depends on SWIM

TBO BENEFITS

Key Benefits of TBO include

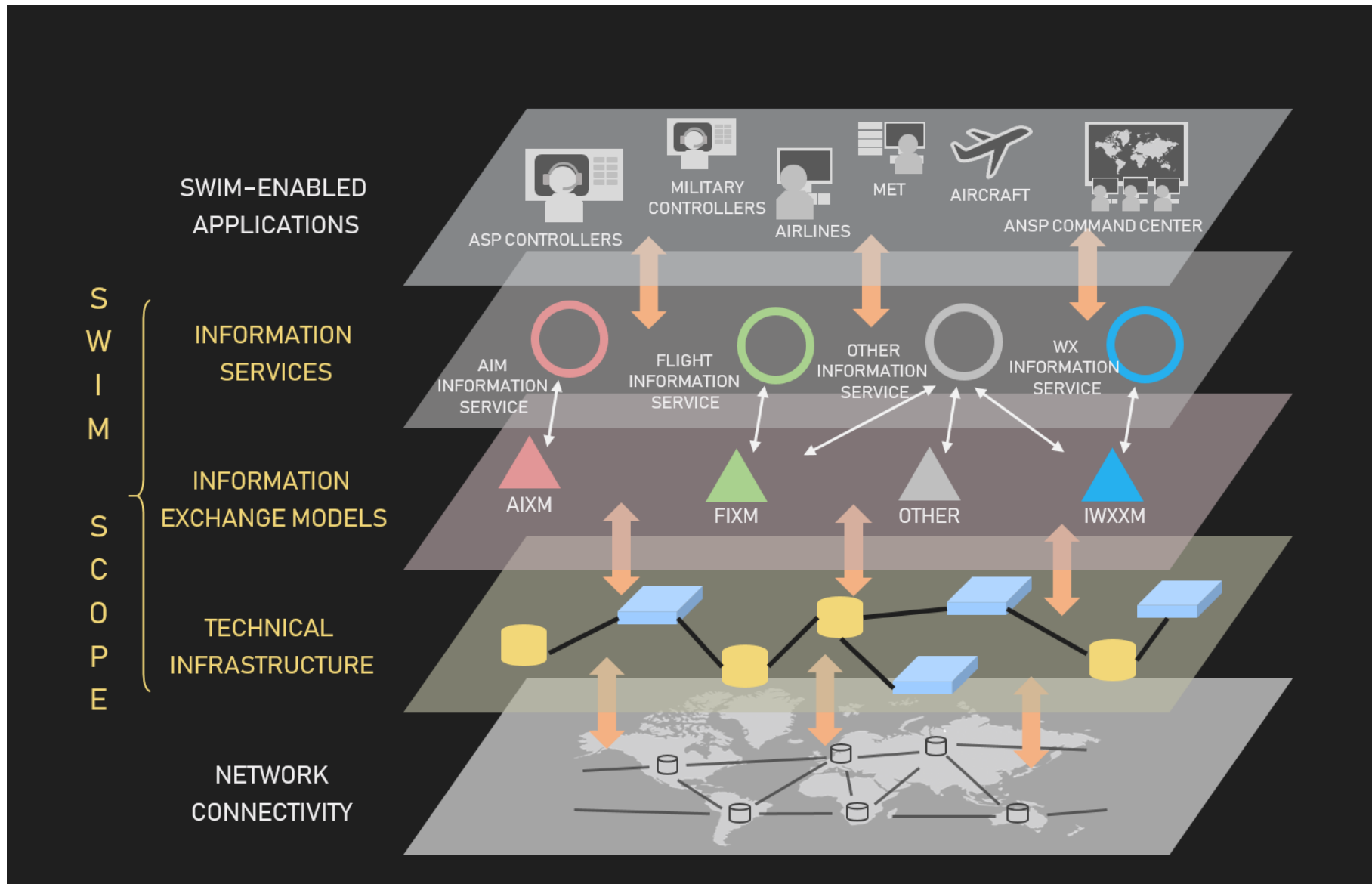
- Increased Airspace Capacity
- Enhanced Predictability
- Improved Efficiency
- Reduced Controller Workload

SWIM STANDARDS



WHAT IS THE SCOPE OF SWIM?

- Whilst closely related, both application and IP network infrastructure aspects are not in the scope of SWIM
- The scope of SWIM is defined by its concept which puts information services at its centre
- Together SWIM information services form a pool that convey standardised information using technical infrastructure capabilities
- The next image illustrates the Scope of SWIM





WHAT DOES IT MEAN TO IMPLEMENT SWIM?

- *“Implementing SWIM” means providing and/or consuming information services*
- *Implementing SWIM is more than implementing a messaging solution only*
- *It is also about understanding the future operational processes and identifying the needed information exchanges in support of collaborative applications*

SERVICE ORIENTATION

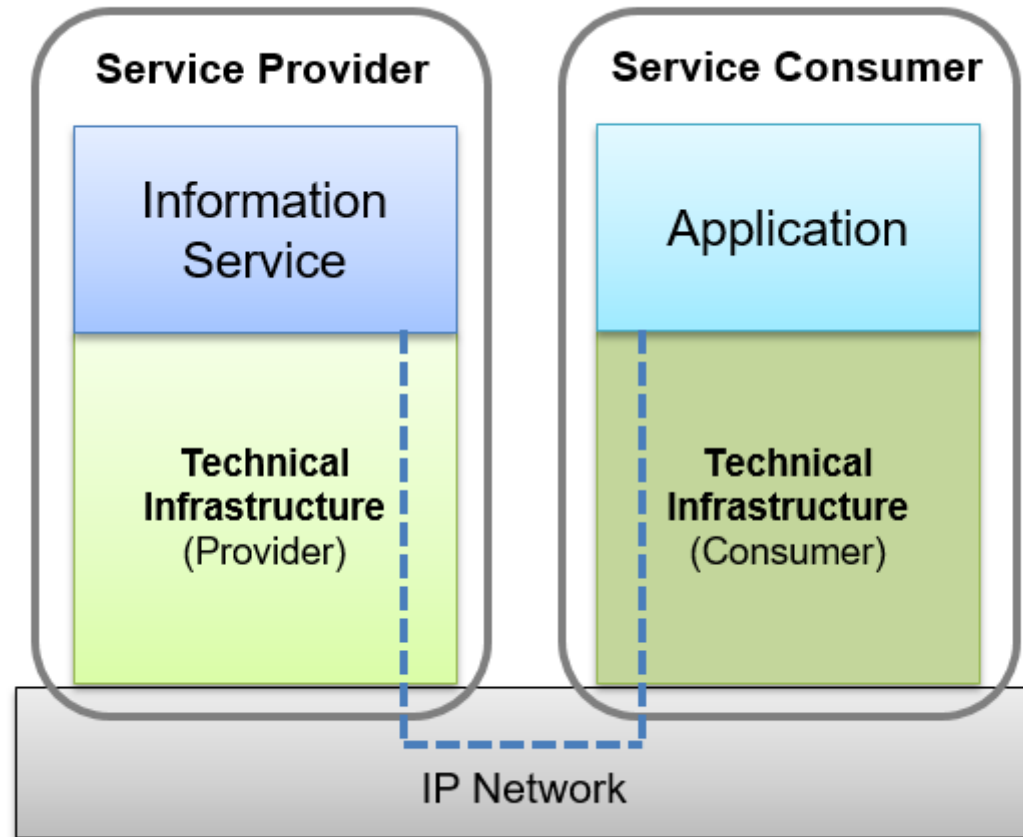
What is Service Orientation?

- *Service orientation is the approach taken in determining the information services*
- *Service orientation brings modularity, reusability and other best practices from the IT world to ATM*
- *There is no formally agreed definition of SOA (service-oriented architecture)*
- *Partitioning of functionality into self-contained and therefore reusable services which can be discovered by potential consumers. That is a key feature of SOA*
- *Existing on-line services may become SWIM information services when fulfilling the requirements of SWIM*



SWIM INFORMATION EXCHANGE

- *SWIM information is exchanged between **SWIM producers** and **SWIM consumers***
- *The exchange follows the appropriate mechanisms available today in SWIM*
- *For example, SWIM TI Profiles and Message Exchange Patterns*
- *SWIM TI Profile specify the interface binding requirements and the infrastructure capability*



COMMON STANDARDS

- The [AIRM](#) (ATM Information Reference Model) is the reference vocabulary for defining air traffic management information.
- Information exchange models such as [AIXM](#), [FIXM](#) and [IWXXM](#) provide syntax and semantics.
- These are elements which are key to allowing interoperability.
- In addition, the [EUROCONTROL SWIM Specifications](#) provide a framework for the development and assessment of SWIM services.



COMMON STANDARDS

Information Exchange Models

- AIXM: <https://aixm.aero/>
- FIXM: <https://fixm.aero/>
- IWXXM: <https://community.wmo.int/iwxxm>



SWIM INFORMATION SERVICES

- SWIM information services are some of the key concepts which make SWIM possible
- Information services are a form of interaction between organisations
- Using Information Services, organisations exchange information (or make it available) with other organisations.
- The Service Provider is an organisation that makes an information service available for use by other organisations
- The Service Consumer is an organisation that decides to use a service to interact with a service provider
- The two organisations exchange information through the Information Service to achieve business goals



EUROPEAN INFORMATION SERVICES

- Examples of Information Services are the EUROCONTROL Network Manager B2B services
- They are best-known European SWIM information services
- NM B2B is used to exchange and share information at the network level with the European Network Manager
- Examples are: Flight Planning and Filing, Slot Allocation, etc

SWIM SERVICE REGISTRIES

- SWIM Service Registries are online directories of information about SWIM Services
- There isn't a single global SWIM Service Registry, there are several, worldwide
- The most well-known one is run by Eurocontrol: <https://eur-registry.swim.aero/services>
- The purpose of the SWIM Service Registries is to allow potential consumers to discover SWIM services
- SWIM producers will publish the services they can provide using the SWIM Registry
- SWIM consumers look up in the SWIM Registry what SWIM services are available and what are the pre-requisites to consume those services.
- Note that producers of SWIM information do not necessarily need to be the ones also providing the service via which information is exchanged



SWIM SERVICE REGISTRIES

- SWIM Service use a formal registration process to store, catalogue and manage information about SWIM services
- The information is used by SWIM consumers to identify which services they are interested in
- A SWIM service listed in a SWIM Registry will have information, for example, about the data format and interfaces that are required
- We can filter the SWIM Services by “Service Provider”, “Service Type”, etc.
- DEMO

Known deadlines for the implementation of SWIM

ICAO GANP AND ASBUS

- The GANP document was published in 2016, and means to cover the expected evolution between 2016 and 2030
- The document is 142 pages long, and covers many technologies
- The area that covers AFTN, AMHS and SWIM is in Roadmap 2
- In 2025, it is easy to see where the predictions have gone wrong, or have not materialised yet
- However, some predictions are correct, and we should review them

GANP R - ROADMAP 2 – BLOCK 0 TIME FRAME (2018)

The following statements are taken from Appendix 5 of GANP 2016

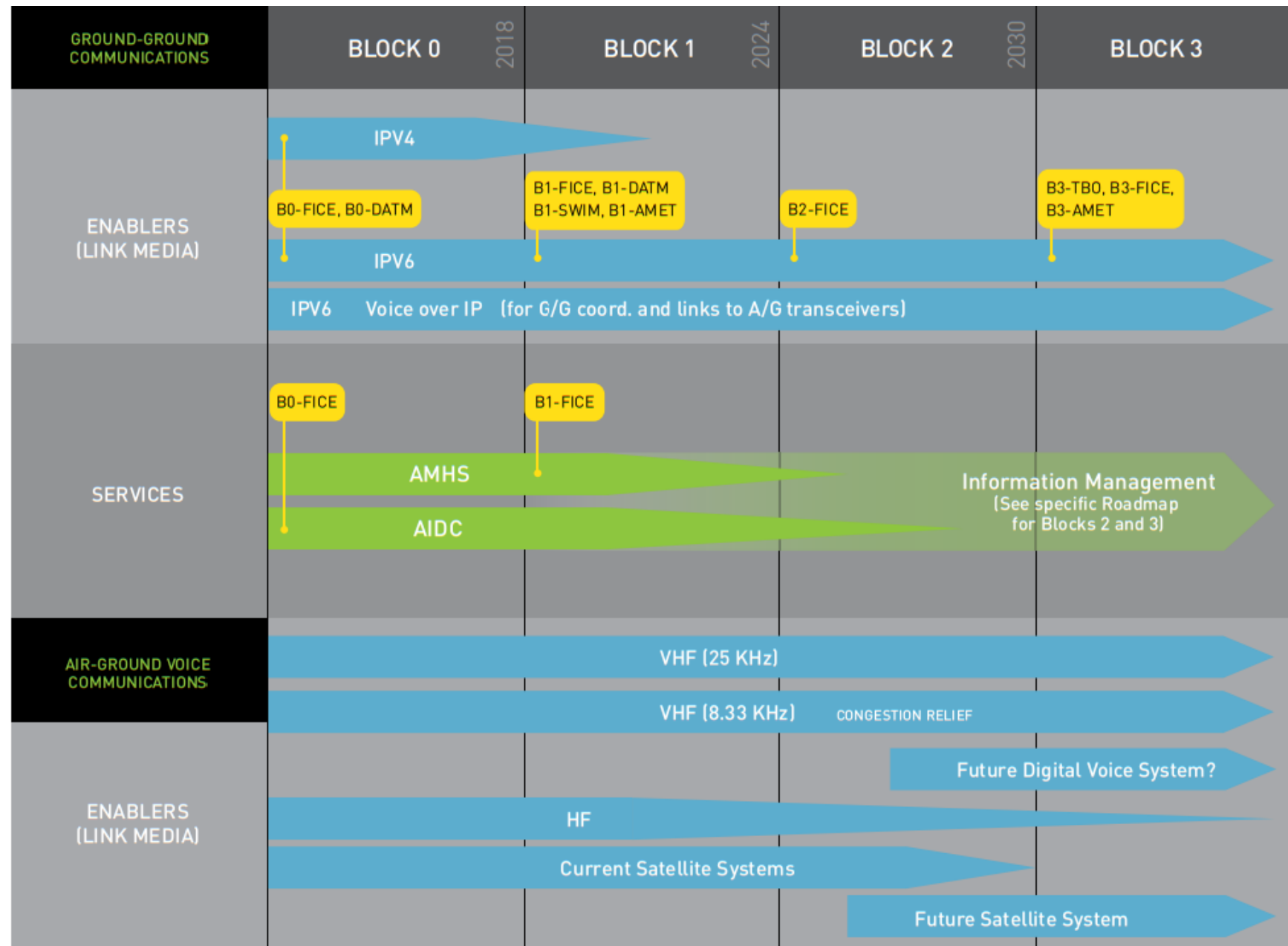
- *IP networks will continue to be deployed. Existing IPV4 systems will be gradually replaced by IPV6*
- *Two major ground-ground communications services will be in operation:*
 - *ATS messaging operating over AFTN and/or AMHS in some areas.*
 - *ATS interfacility data communications (AIDC) for flight coordination and transfer.*
- *ATS messaging is used worldwide for the communication of flight plans, MET, NOTAMS, etc. over AFTN technology.*
- *Migration towards AMHS (directory, store and forward services) over IP (or using ATN in some regions) will progress in all regions.*
- *The beginnings of SWIM will start to appear. Operational services will be offered by some SWIM pioneer implementations over IP, surveillance data distribution and MET data will also be distributed over IP.*
- *Migration to digital NOTAM has started in the United States and in Europe.*

GNAP R - ROADMAP 2 – BLOCK 1 TIME FRAME (2024)

More predictions

- *Digital NOTAM and MET (using the AIXM and IWXXM data exchange formats) will be widely implemented over IP networks.*
- *FIXM will be introduced as the global standard for exchanging flight data prior to (Block 1) and during the flight (Block 2).*
- *ATS messaging will migrate to AMHS supported by directory facilities that will include common security management by the end of Block 1.*
- *AIDC services will fully migrate towards using IP networks.*
- *Initial 4D air-ground services will require ground-ground inter-centre trajectory and clearance coordination via AIDC extensions or new flight data exchanges compatible with the SWIM framework.*
- *SWIM SOA services will mature and expand publish/subscribe and request/reply services in parallel to the more traditional messaging services based on AMHS but both will use the IP network.*

ICAO GANP 2016 Roadmap 2





THE TRANSITION FROM AFTN TO AMHS

- As we could see, the predictions of ICAO are always optimistic when it comes to the time frames
- AFTN is still being used, and AMHS is still growing, not being replaced by SWIM
- There are newer versions of that diagram with later dates, but still very optimistic

THE ICAO MET SWIM ROADMAP – MARCH 2025

- Out of MET, AIS and FPL, the MET domain is the one that is most advanced in SWIM / XML usage
- Published in March 2025, the new ICAO MET SWIM Roadmap is more realistic than the GANP
- It considers that up until the end of 2024, MET messages were being sent via AMHS FTBP
- The Capabilities Expected During Block 2 (2025-2030) are AMHS FTBP and (Optional) AMQP/HTTP
- Only after 2037 does AMQP/HTTP become the only capability



ICAO MET SWIM Roadmap (v3 – March 2025)

	Capabilities Expected During Block 1 (2019-2024)	Capabilities Expected During Block 2 (2025-2030)	Capabilities Expected During Block 3 (2031-2036)	Capabilities Expected During Block 4 (2037+)
Communication Protocols	AMHS FTBP	AMHS FTBP AMQP/HTTP (optional)	AMHS FTBP AMQP/HTTP	AMQP/HTTP
Information Exchange Services	RODB TAC request/reply RODB IWXXM request/reply	RODB TAC request/reply RODB IWXXM request/reply OGC Standardized Services (optional)	RODB IWXXM request/reply OGC Standardized Services	OGC Standardized Services
Data Addressing	AFS Addressing	AFS Addressing IP (optional) SWIM Registry (optional)	AFS Addressing IP SWIM Registry	IP SWIM Registry
Information Exchange Flow	NOC, ROC, RODB, IROG	NOC, ROC, RODB, IROG Dynamic (optional)	NOC, ROC, RODB, IROG Dynamic	Dynamic
Data Aggregator	NOC, ROC, RODB, IROG	NOC, ROC, RODB, IROG SWIM Aggregator (optional)	NOC, ROC, RODB, IROG SWIM Aggregator	SWIM Aggregator

The status of SWIM around the world

IMPLEMENTATION OF SWIM

- There are currently several ongoing implementations of SWIM services worldwide
- In the USA, SWIM is part of NextGen
- In the European Union, SWIM is part of SESAR (Single European Sky ATM Research)
- There are other programmes ongoing in Asia, Africa and South America
- The implementation of SWIM is a continuous process that occurs at regional and local levels
- SWIM is a key component of the ICAO Global Air Navigation Plan (GANP)
- They are part of the Aviation System Block Upgrades (ASBUs), B1-SWIM and B2-SWIM

The transition to SWIM

THE TRANSITION FROM AFTN TO AMHS

- As we could see, the predictions of ICAO are always optimistic when it comes to the time frames
- AFTN is still being used, and AMHS is still growing, not being replaced by SWIM
- ICAO tried to make the transition from AFTN to AMHS easier, for example, deciding that:
 - The actual payload of a message in AFTN remained the same in AMHS
 - There were two versions of AMHS: Basic and Extended
 - Basic AMHS was very similar to AFTN
 - Extended AMHS was optional
 - ICAO created a series of documents that specified the conversion of AFTN messages to AMHS and vice-versa
 - Those documents, ICAO Doc 9880, are still being maintained

THE TRANSITION FROM AFTN AND AMHS TO SWIM

- For the transition of AFTN and AMHS to SWIM, ICAO has not done as much as it did from AFTN to AMHS
- Differences to bear in mind:
 - In SWIM information is not sent from a “From” originator to a set of “To” recipients
 - In SWIM information is published
 - The actual payload of a message changes radically from text in AFTN/AMHS to XML encoding in SWIM
 - There are few documents that cover the conversion of the payload from existing formats to SWIM formats
 - The conversion of the formats is discouraged
 - Users are encouraged to create and process XML content natively, not via conversion
 - There is no AMHS to SWIM Gateway standard created by ICAO
 - The SWIM documents hardly ever mention the existence of AFTN or AMHS

THE TRANSITION FROM AFTN AND AMHS TO SWIM

- Probably because the differences that exist between AMHS and SWIM are so big, ICAO did not plan for the co-existence of these technologies
- Another reason may be that, considering the length it took AMHS to replace AFTN, ICAO did not want AMHS to co-exist with SWIM for long
- In any case, the lack of a transition organised by ICAO was perceived as a problem
- ANSPs wanted a guidance from ICAO or from manufacturers of ATM software
- We will examine the differences in more detail

DIFFERENCES: ADDRESSING

- An AFTN message has an originator and up to 21 recipients
- An AMHS message has an originator and up to 32k recipients
- In SWIM, a message is published. Whoever is interested, will subscribe and receive the information
- The originator of the information doesn't know who the intended recipients are
- The originator doesn't know (or care) if the recipients received the information

DIFFERENCES: PROTOCOLS

- AFTN uses its own proprietary protocol, based in serial communications, not used by anyone else
- AMHS is based on X.400, which originally used OSI stack but was then adapted to use IP
- AMHS uses X.400 P1, P3 and P7, protocols that are not popular, but at least still in use by EDI, Government, etc.
- SWIM uses web-based protocols, like HTTP, and AMQP
- Web-based protocols are very popular and have excellent support in the developer community nowadays
- The most similar protocol to AFTN or AMHS is AMQP, as messages have subjects, recipients, etc.

DIFFERENCES: CONTENT

- AFTN uses text-based content, with a reduced character set
- AMHS can use the same text-based content than AFTN, but also different types, like binary or General Text
- The important part is that all AFTN content can be conveyed exactly as it is by AMHS
- The payload of, for example, a Flight Plan, doesn't change is sent via AFTN or AMHS
- SWIM uses mostly XML-encoded content, like IWXXM, AXIM or FIXM.
- The content of a Flight Plan is definitely not the same in AFTN-AMHS than in SWIM (FIXM/FF-ICE)
- Since the SWIM content contains more information than the AMHS content, the conversion is not perfect
- From SWIM to AMHS, information will be lost.
- From AMHS to SWIM, important information will not be present



THE MIXED ENVIRONMENT

- SWIM will eventually replace AFTN and AMHS
- There is no plan to change the entire world from AFTN and AMHS to SWIM in a single point in time
- Such a change would be almost impossible to achieve worldwide, at the same time
- The transition will take time, and will be undertaken by different countries at different speeds
- This approach will lead to a mixed environment, with both AFTN-AMHS and SWIM co-existing

European Strategy for the transition to SWIM

EUROPEAN STRATEGY FOR THE TRANSITION FROM AFS TO SWIM

- CP1 Regulation (EU 2021/116) → AF5 SWIM
- Coexistence of AFS (AMHS, AFTN/IP and even AFTN (asynchronous)) with SWIM “native” users
- Progressive transition from AFS to SWIM, this transition takes some time and no impact on day-to-day operations shall be ensured
 - There is a real need to exchange information between SWIM “native” users and AFS users (AMHS, AFTN)
 - AFS users will migrate to SWIM at different paces
 - Different mechanisms are available (AMHS/SWIM Gateway, dual-stack systems)

EUROPEAN STRATEGY FOR THE TRANSITION FROM AFS TO SWIM

- Users impacted by CP1 Regulation (EU 2021/116) → AF5 SWIM (<https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R0116>)
 - Family 5.1.1 Common infrastructure components
 - SWIM Registry
 - Common Public Key Infrastructure (PKI)
 - Family 5.1.2 SWIM Yellow Profile Technical Infrastructure
 - Family 5.1.3 Aeronautical Information Exchange (AIX)
 - ARES (Airspace Reservation/Restriction)
 - ASM (Airspace Management) → enhances FUA

EUROPEAN STRATEGY FOR THE TRANSITION FROM AFS TO SWIM

- Users impacted by CP1 Regulation (EU 2021/116) → AF5 SWIM (<https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R0116>)
 - Family 5.1.4 MET Information Exchange (IWXX)
 - Volcanic ash concentration
 - MET information supporting aerodrome procedures
 - MET information supporting ENR/APP procedures
 - MET information supporting network information management
 - Family 5.1.5 Cooperative NW Information Exchange (Network Manager)
 - Family 5.1.6 Flight Information Exchange using YP (FIX)
 - FF-ICE/R1 (flight plans (route validation, 4D trajectory, flight status, etc.)
 - FUM (Flight Update Messages) / DPI (Departure Information)

The AST-TF

THE AST-TF

- Given the lack of guidance from ICAO for the transition from AMHS to SWIM there was little activity in this field
- The AFSG was the “AERONAUTICAL FIXED SERVICES GROUP”
- It mostly dealt with AMHS issues and standards
- It was composed of a Planning Group (PG) and an Operations Group (OG)
- I was a member of the AFSG Planning Group
- In 2024 the group was renamed from AFSG to AST-TF (AFS TO SWIM TRANSITION TASK FORCE)
- I am now a member of the AST-TF Planning Group

3) European WGs working on this transition (ICAO, Eurocontrol NM, etc.)

ICAO



EUROPEAN AVIATION SYSTEM PLANNING GROUP



AMHS To SWIM Transition Task Force (AST TF)

AMHS To SWIM Transition **Planning Group (AST PG)**

AMHS To SWIM Transition **Operations Group (AST OG)**

3) European WGs working on this transition (ICAO, Eurocontrol NM, etc.)

ICAO



AMHS To SWIM Transition Task Force (AST TF) Work Programme

OG:

Network issues
COM Centers coordination
3rd party AMHS GWs
Documentation Maintenance
Interregional extended AMHS
Use of local DSA

PG:

AMHS Implementation issues
TCP/IP WAN requirements
OPMET Digital Data Implementation
ATN
SWIM & Transition to SWIM
Security

3) European WGs working on this transition (ICAO, Eurocontrol NM, etc.)



SUPPORTING
EUROPEAN
AVIATION

Information Management Team (IMT)

SWIM

System-wide information management

We help manage ATM-related information and its exchange between qualified parties through interoperable services.

4) Activities by the IMP (Information Management Panel) and proposed coordination (e.g., ICAO Doc. 10203)

Term of Reference

Background	The Information Management Panel (IMP) is to be established to develop a global and harmonised interoperable approach and elaborate on necessary concepts in order to ensure effective management of Information, including identifying the need for new information exchange formats, on a system-wide basis within the Air Navigation System. A global approach on Information Management is essential to ensure global interoperability and standardisation across all data domains and to support activities such as FF-ICE, the evolution of MET towards digital information exchange and a NOTAM system review.
Scope	The Information Management Panel will investigate and develop solutions supporting the planning framework on Information Management contained in the Global Air Navigation Plan, including further development of SWIM using as a basis the SWIM Concept as elaborated by the ATMRPP. The IMP will develop a global interoperability framework for international air navigation. Its components (e.g. technical resources such as information models and associated exchange formats, service models, governance functions and structure etc...) will be worked upon as they are identified and agreed during the course of the IMP proceedings.
Required	The panel shall be preferably composed of experts involved:

ICAO Doc. 10039
Manual on SWIM
Mgmt. concept

ICAO Doc. 10203
Manual on SWIM
Implementation

4) Activities by the IMP (Information Management Panel) and proposed coordination (e.g., ICAO Doc. 10203)



IP TF



Information Management Panel (IMP)

AMHS To SWIM Transition Task Force (AST TF)

- ✓ **Ensure a smooth transition from AFS to SWIM**
- ✓ **Make sure there is no negative impact on AFS while transitioning to SWIM**
- ✓ **Collaborate with IMP to enhance ICAO Docs. for the transition**

THE CHALLENGE: TRANSITION TO SWIM

- As noted, the differences between SWIM and AMHS are vast. They are even more so with AFTN
- How can we organise the transition from, for example, Flight Planning to FF-ICE?
- The systems that claim to have FF-ICE operational at the moment, are doing so with the so called dual-stack
- Using the Dual-Stack approach they accept and generate both ICAO Doc 4444 FPLs and FF-ICE
- One example using the Dual-Stack approach is Eurocontrol's NM
- Since they can process and generate both message formats, they can connect simultaneously to both countries that use ICAO Doc 4444 FPLs (probably all of them, as of December 2025) or anyone using FF-ICE.



THE CHALLENGE: DUAL STACK

- The Dual-Stack approach is key in maintaining interoperability during the transition to SWIM
- This prevents the need for hard cut-over from AMHS to SWIM
- It allows different countries to migrate when they are ready, not all at the same time
- The disadvantage is that the Dual-Stack system has at least double the complexity of the AMHS system or the SWIM system, as it needs to appear native to both systems
- NOTE: Share the Dual Stack slides



THE CHALLENGE: TRANSITION TO SWIM

- Note that, even if a country in Europe is claiming to connect to Eurocontrol's NM using FF-ICE, that doesn't mean that their national ATM system is actually processing those FF-ICE messages.
- It may mean that they have an application that is able to connect and get those FF-ICE messages

The need of an AMHS/SWIM Gateway in the
transition to SWIM

THE AMHS/SWIM GATEWAY

- ICAO did not create a specification for an AMHS/SWIM Gateway, as it did for the AFTN/AMHS Gateway
- The lack of this specification made many ANSPs wonder how they should organise the transition
- From the beginning, it was clear that an AMHS/SWIM Gateway would not be like an AFTN/AMHS Gateway
- An AFTN/AMHS Gateway enables sufficient communication between AFTN and AMHS
- Only AMHS messages like the ones with FTBP cannot be converted to AFTN
- For the case of the AMHS/SWIM Gateway, this was never the objective

THE AMHS/SWIM GATEWAY

- The objective of an AMHS/SWIM Gateway Specification was to help in the transition to SWIM
- It was not the objective of AMHS/SWIM Gateway Specification to delay or prevent the transition to SWIM
- Due to the complexity of the X.400 technology, the data currently in AMHS format is hard to access
- It was the hope that an AMHS/SWIM Gateway Specification would help by moving the X.400 P22 encoded data to a technology aligned with SWIM, for example, as a payload to an AMQP message.
- It would be then much easier for a SWIM Native system to process AMHS data, without having to need, for example, an X.400 MTA and tools to encode and decode X.400 P22 content and implement P1 protocols.

The SWAMWAY Study Group

SWAMWAY SG

- Due to the lack of initiative by ICAO or by any regional ICAO group, several companies decided to take action
- This led to the creation of an AMHS/SWIM Gateway Study Group
- The group discussed for a long time how the transition to SWIM could be achieved
- The group decided to explore the creation of a specification of an AMHS/SWIM Gateway
- This group became known as the SWAMWAY Study Group, or SWAMWAY SG



SWAMWAY PARTICIPANTS

- The following companies are part of SWAMWAY
 - Copperchase, Frequentis-Comsoft, Indra-Avitech, Telefónica and Thales
- The following ANSPs are part of SWAMWAY
 - Austro Control (Austria) and Enaire (Spain)

SWAMWAY GOALS

- The goals of SWAMWAY was to create and maintain an AMHS/SWIM Gateway Specification
- It was not to produce a product to sell
- The companies involved did not receive any funding or payment to produce the Specification
- The specification is open source, i.e. freely available
- SWAMWAY doesn't recommend any specific way to implement the AMHS/SWIM Gateway
- Another goal was to ensure interoperability between different providers of AMHS/SWIM Gateway products

SWAMWAY SOURCES

- The companies and ANSPs that are members of SWAMWAY have a lot of expertise in both AMHS and SWIM
- Several other experts were consulted, worldwide, and were invited to participate in SWAMWAY
- All ICAO Regional Offices were also contacted and were asked to provide feedback
- Some experts did not join SWAMWAY, but they provided their feedback

SWAMWAY REFERENCES

- For AMHS, the reference was ICAO Doc 9880 Edition 2, as Edition 3 had not been published then
- For SWIM, the reference was SWIM Technical Infrastructure Yellow Profile by EUROCONTROL and the AMQP standards by the OASIS group
- Before starting to write the specification, it was decided that it should be as close as possible to the AFTN/AMHS Gateway Specification
- This similarity makes it much easier to developers that have implemented an AFTN/AMHS Gateway to implement an AMHS/SWIM Gateway
- The similarity would also avoid all the revisions that took place over the years until Doc 9880 was published
- SWAMWAY produced a single document: the AMHS/SWIM Gateway Specification

SWAMWAY's Specification

EDITORIAL PRACTICES

- ASG uses the same editorial practices as the AFTN/AMHS Gateway defined in ICAO Doc 9880
- The specifications that include the verb “**shall**” are essential to be implemented to secure proper operation of the AMHS/SWIM Gateway.
- The specifications that include the verb “**should**” are recommended for implementation in the AMHS/SWIM Gateway.
- The specifications that include the verb “**may**” are optional.

POSSIBLE PROTOCOLS TO USE FOR SWIM

- For AMHS, the choice of protocols was clear: a Gateway application was needed with X.400 P1 to use externally
- The option to define at a user level, i.e. using X.400 P3 and X.400 P7 was discussed but not accepted
- For SWIM, there was a choice of protocols
- SWIM information is exchanged between SWIM producers and SWIM consumers following the appropriate mechanisms available today in SWIM (e.g., SWIM TI Profiles and Message Exchange Patterns)
- SWIM TI specifies interface binding requirements and infrastructure capabilities
- They depend on the specific SWIM TI Profile SWIM users utilize to exchange information.

POSSIBLE PROTOCOLS TO USE FOR SWIM

- Among the different SWIM profiles published by EUROCONTROL, the SWIM Yellow profile is the best suited based on the type of information that the AMHS/SWIM Gateway exchanges
- The SWIM Yellow Profile offers four different types of service interface bindings: WS Light, WS SOAP, WS-N SOAP and AMQP messaging.
- Analysis of the bindings showed that AMQP is the best in terms of performance and security
- AMQP stands for “Advanced Message Queuing Protocol”
- AMQP offers interoperability, that is the key concept of the SWIM paradigm

THE CREATION OF THE SPECIFICATION

- The creation of the ASG Specification took about two years and dozens of meetings
- As we were writing the Specification, technical problems emerged that had to be solved
- SWAMWAY consulted with the ATS TF Planning Group through the process
- Among them was the issue of the level of “awareness” of AMHS that SWIM AMQP messages should have
- Note that the ASG Specification doesn’t provide a manual on how to create an ASG, just a set of requirements

AMHS-AWARE AND AMHS-UNAWARE

- The most difficult concepts to understand in the ASG Specification are the “AMHS-Aware” and “AMHS-Unaware”
- This is referred to as two service levels
- A SWIM message, sent via AMQP is normally “AMHS-Unaware”, that is, it doesn’t know anything about AMHS
- If the sender of a SWIM message, to be sent via AMQP wants to address a particular AMHS user, it has to add some optional fields in the AMQP message. This makes the AMQP message “AMHS-Aware”
- The fields are, for example, the address of the originator and the recipients to use, the ATS Priority, etc

AMHS-AWARE AND AMHS-UNAWARE

- Bear in mind that the vast majority of AMQP messages created will be “AMHS-Unaware”
- Only a very small subset, created by SWIM Producers that are aware of AMHS will be “AMHS-Aware”
- The AMHS/SWIM Gateway shall convert both “AMHS-Unaware” and “AMHS-Aware” AMQP messages
- We will look at this subject in more detail when we cover the conversion in the ASG Specification

THE ADOPTION BY EUROCONTROL AS EUR DOC 047

- Once the ASG Specification was written, it was presented by SWAMWAY to the AST-TF Planning Group
- The AST-TF Planning Group approved of the AMHS/SWIM Gateway Specification
- The AST-TF Planning Group recommended that the AST-TF parent group endorsed the Specification
- Eventually it was adopted by Eurocontrol, and given a document reference of EUR Doc 047
- Note that this could eventually be adopted by ICAO with a different reference
- The endorsement of the Specification by Eurocontrol is a much faster process than by ICAO

THE ADOPTION BY EUROCONTROL AS EUR DOC 047

- ICAO EUR Doc. 047 approved by EASPG since the end of 2023
- Maintained by AST TF PG under AST TF mandate
 - Improve wording and readability
 - Include additional topics based on discussions with different bodies and organizations (e.g., AMHS/SWIM Gateway test plan)
- Public and available at [ICAO website here](#)
- ICAO EUR Doc 047 has been used by different organizations for their own use cases (ENAV, DFS, ENAIRE, ...)

What to expect from an AMHS/SWIM Gateway

WHAT TO EXPECT FROM AN AMHS/SWIM GATEWAY

- The expectation of an end user of the AMHS/SWIM Gateway needs to be adjusted
- For example, the AMHS/SWIM Gateway **won't perform any content conversion**
- If an AMHS user sends a METAR message to the AMHS/SWIM Gateway, it won't be converted to IWXXM, but it will be converted to an AMQP message with the content of the METAR
- If a SWIM Service provides IWXXM data via AMQP, an AMHS/SWIM Gateway may convert it to an AMHS message that contains the IWXXM data, but it won't convert it to, for example, a METAR

WHAT TO EXPECT FROM AN AMHS/SWIM GATEWAY

- The AMHS/SWIM Gateway can help in the transition to SWIM, but it is not a tool to convert AMHS to SWIM
- For example, if an information provided decides to stop sending information via AMHS but it now starts to publish it in SWIM, the AMHS/SWIM Gateway could be used to keep the information being sent to AMHS users.
- We will examine current Use Cases for the AMHS/SWIM Gateway in Module 5 of this Training
- More Use Cases are expected to emerge, once more SWIM services are operational
- We cannot envisage all the possible uses of an AMHS/SWIM Gateway right now

Demo of an AMHS/SWIM Gateway

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